

VISHAAL CHANDRASEKAR

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PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence Engineer with a Master's in AI and a track record of designing and deploying scalable AI/ML solutions across cloud platforms. Expertise in NLP, generative AI, and MLOps, with a focus on ethical AI practices. Adept at translating complex technical concepts into actionable insights, leading cross-functional teams, and delivering robust production-ready systems

PROFESSIONAL EXPERIENCE

Machine Learning Software Engineer, Contraca | Boston, MA

January 2025 – Present

- Developed custom NLP solutions for entity recognition and document classification using AWS Comprehend, leading to improved data accuracy and streamlined ML pipelines for Responsible AI applications.
- Engineered a scalable classification system on AWS using SageMaker, S3, Lambda, DynamoDB, and EC2, enhancing deployment speed by 30% and ensuring high reliability.
- Drove cross-functional collaboration by clearly communicating technical concepts and project milestones with business goals.

Artificial Intelligence Engineer, Reframe Solutions | Hartford, CT

May 2023 – December 2023

- Led the development of a GPT-2 based generative AI infrastructure for a government business registration platform, enabling 1,000+ live online registrations while upholding Responsible AI practices.
- Enhanced data workflows from Azure to SQL Server, reducing processing time by 15% through optimization of AI-driven processes.
- Upgraded GPT-3.5 for predictive financial analysis by integrating Perception algorithms and synthetic data via diffusion models, achieving a 10x performance improvement in risk mitigation.

Data Scientist Intern, Mastech Infotrellis | Chennai, India

February 2022 – July 2022

- Conducted deep data analysis on healthcare and consumer datasets using Python, Power BI, IBM Cognos, and Tableau, boosting predictive accuracy by 20%.
- Created a robust Python ML codebase for data preprocessing, feature engineering, and classification (Logistic Regression, Random Forest), improving decision-making for unstructured data.
- Partnered with cross-functional teams to design and implement data-driven solutions that addressed complex business challenges.

EDUCATION

University of Bridgeport, School of Engineering

Master of Science in Artificial Intelligence, Concentration in Computer Technology

August 2022 – December 2024

Cumulative GPA: 3.67/4.00

SRM University, SRM Institute of Science and Technology

Bachelor of technology in Computer Science and Engineering

June 2018 – May 2022

Cumulative GPA: 3.74/4.00

TECHNICAL SKILLS

Languages: Python, R, C++, SQL (MySQL), JavaScript, TypeScript, HTML, CSS

Frameworks: Lang Chain, Hugging face, PyTorch, OpenCV, Transformers, BERT, LLAMA, CNN, RNN, GAN, RAG, Spark

DevOps and API tools: AWS (SageMaker, EC2, Lambda, DynamoDB, S3), Oracle Cloud, Linux, Git, Hadoop, DynamoDB

Architecture: LLMs (LLAMA 2, GPT 3), Kernels, ML Compilers, GPU Parallel Processing, ETL

Tech Certification: Prompt Engineering, Azure OpenAI Application, Data Science, Oracle Infrastructure

RESEARCH EXPERIENCE

Ensuring Secure Children's Bedtime Stories: Controlling Large Language Models against Adversarial Attacks

Ausif Mahmood, Vishaal Chandrasekar | Bridgeport, CT

December 2023

Designed a robust framework incorporating diffusion models with security breakpoints to address vulnerabilities and mitigate risks associated with RLHF on Llama 3 models in post training and improved the trustworthiness of adversarial AI with control mechanisms to enhance the safety and reliability of generative AI.

Data Pedigree Analysis: Lineage Analysis

Vishaal Chandrasekar | Remote

November 2024

Researched data redundancy and provenance, analyzing its impact on Perception algorithm training, performance optimization, and ethical development through advanced preprocessing techniques and applied mathematical foundations in pre-training models.

PUBLICATIONS

Face Detection and Image Recognition with python - Vishaal Chandrasekar. *International Journal of Science and Research (IJSR), Volume 10, Issue 3*

March 2021

Researched and implemented face detection and recognition using Python and OpenCV leveraging the haar Cascade algorithm for detection and the local binary pattern histogram algorithm for recognition. Explored the multifaceted application of artificial intelligence and machine learning, showcasing their roles in advanced computer vision, image analysis, and edge detection technology